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COURSE NAME : Design

Analysis of Algorithm

CO5 – AT2 --Pseudocode Writing Task (Algorithm Design)

Q20. Bin Packing Problem – Backtracking Algorithm

1. Problem Understanding

The Bin Packing Problem aims to place all items into a minimum number of bins such that the total size of items in each bin does not exceed its capacity.

Input

- items[n] – sizes of n items
- capacity – maximum capacity of each bin

Output

- binAssignment[n] – bin number assigned to each item
- Minimum number of bins required

Constraint For every bin:

Total size of items $\leq$ Bin Capacity

2. Backtracking Idea

The algorithm places items one by one into available bins.

- Try placing the current item in each existing bin.
- Check whether the item fits.
- If it fits, place it and recursively process the next item.
- If the placement fails later, remove the item (backtrack).
- A new bin is opened only when necessary.

- Pruning avoids exploring bins where the item cannot fit. This follows the systematic backtracking style used in the uploaded assignment.

3. ALGORITHM

BinPacking(items, n, capacity)

1. Initialize binLoad[] = 0 and binAssignment[].
2. FUNCTION Backtrack(index, bins)
3. IF index = n THEN return bins.
4. FOR each bin from 0 to bins-1 DO
5. IF binLoad[bin] + items[index] $\leq$ capacity THEN
6. Place item[index] in bin and update binLoad.
7. Backtrack(index + 1, bins).
8. Remove item[index] from bin. // Backtrack
9. IF no existing bin works, open a new bin.
10. Return the assignment using the minimum number of bins.

4. Pruning Conditions

1. Capacity Check:

If binLoad[bin] + items[index] > capacity, the bin is not explored.

2. Best Solution Check:

If the current number of bins is already greater than or equal to the best solution, further exploration can be stopped.

3. Backtracking:

After trying a placement, the item is removed so another possible assignment can be explored.

5. Example

Items: {4, 3, 5, 2}

Bin Capacity: 7

Possible assignment:

Bin	Items	Total
Bin 1	4 + 3	7
Bin 2	5 + 2	7

Output:

1. Minimum number of bins = 2
2. Bin 1 $\rightarrow$ Items 4, 3
3. Bin 2 $\rightarrow$ Items 5, 2

6. Correctness

- Every item is considered.
- An item is placed only when it does not exceed bin capacity.
- Backtracking explores alternative placements.
- Invalid assignments are pruned early.
- Therefore, all feasible assignments can be explored and the minimum number of bins can be identified.

7. Complexity

- **Time Complexity:** $O(m^n)$ in the worst case, where m is the number of bins considered.
- **Space Complexity:** $O(n)$ for bin assignments, bin loads, and recursion stack.

8. Result

The backtracking algorithm systematically assigns each item to a suitable bin, prunes infeasible placements, and backtracks whenever a placement cannot lead to a valid solution. It can therefore find an optimal bin assignment for small problem instances.